

Short communication

Errors in velocities and displacements deduced from accelerographs: An approach based on the theory of error propagation

Stathis C. Stiros

Geodesy Laboratory, Department of Civil Engineering, Patras University, Patras 26500, Greece

Received 10 January 2007; received in revised form 6 July 2007; accepted 9 July 2007

Abstract

Based on the theory of errors, and in particular on the law of error propagation and approximation techniques, we present some simple formulae for random errors of velocities and displacements computed on the basis of numerical integration of accelerometer records. These errors are regarded as function of the characteristic errors of accelerometers and of the duration of the signal for velocities, and of the square of the duration of the signal for displacements. Instabilities in the sampling rate introduce additional noise, which is proportional to the sum of the squares of measurements of acceleration, mostly influenced by high peaks. Because of uneven weighting, however, peaks at early stages of the record are more important than in their later stages.

© 2007 Elsevier Ltd. All rights reserved.

Keywords: Accelerograph; Acceleration; Velocity; Displacement; Error; Propagation; Sampling; Time series

1. Introduction

Acceleration, velocity and displacement at a point of a continuous medium are connected with simple relationships—a simple or double integration, respectively. For this reason accelerometers, which can measure acceleration with very high accuracy, have been used to compute the earthquake-induced velocity and displacement histories of the ground or of certain structures during earthquakes [1,2] (Fig. 1).

Furthermore, in the mid-1980s there have been efforts at producing accelerometer-based systems for the computation of coordinates and the navigation of vessels and airplanes (inertial positioning systems or inertial survey systems). Soon, however, it was realized that the calculated coordinates of the moving crafts were contaminated by serious errors leading to unrealistic apparent displacement (“drifts”, e.g. [3]). For this reason inertial positioning systems were abandoned, and only recently have been used as sensors subordinate to GPS; i.e. to record vessel track in short periods of GPS outages [4–6].

Although it was clear that the drift error was serious in the case of relatively long records of accelerometer-computed displacement, much shorter signals, such as those produced by earthquakes also proved to be contaminated by this type of error. A characteristic example is shown in Fig. 2, revealing that the drift in a strong earthquake may exceed the value of 1 m ([7]; see also [3]).

The reasons for such errors are not fully understood, mainly because this subject has been studied by very few investigators. For instance, Trifunac and Lee [8] focused on the assessment of accelerometer records through a comparison with transducer-based displacements; Thong et al. [9,10] analyzed accelerograph errors as function of time and of sampling frequency using signal analysis techniques; Trifunac and Todorovka [11] summarized the limitations in reconstructing the displacement histories based on accelerographs and highlighted the quadratic form of apparent displacements, to a major degree assigned to major peaks in the accelerograph records; Hu and Ma [12] found that for big angular vibration the frequency characteristic error is the main source of error. On the other hand, various authors have proposed techniques to correct the drift (for a summary and a recent approach see [3]).

E-mail address: stiros@upatras.gr

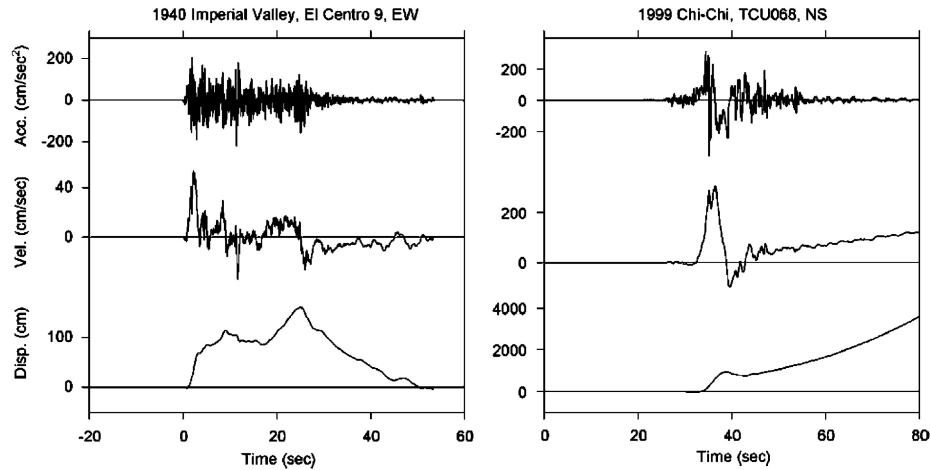


Fig. 1. Acceleration, velocity and displacement for two different earthquakes (after [2]).

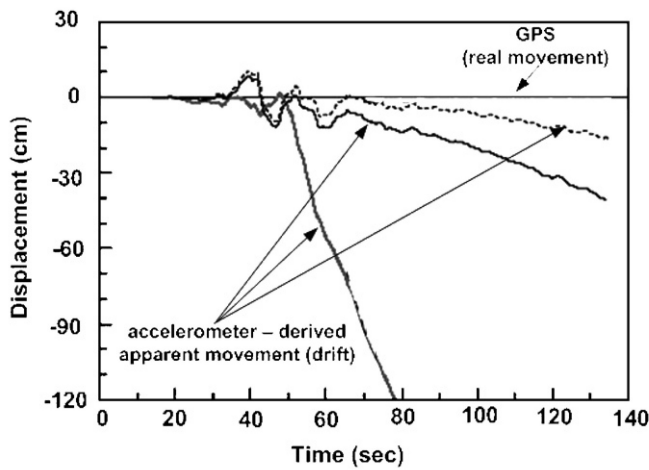


Fig. 2. Drift of three different accelerometers for the 1999 Chi-Chi (Taiwan) earthquake in comparison with GPS data (based on [7]).

Still, the questions arising are (1) why errors are more important in displacements than in velocities? (2) why errors seem to increase with the length of time series (see Fig. 2)? and (3) is it possible to derive certain simple formulas that relate the errors in measurements with the errors in estimations of velocities and displacements?

In this article, we try to present a theoretical answer to these problems based on the theory of errors, especially the law of propagation of errors [13] and approximation techniques.

2. Methodology

We assume that a certain accelerometer has recorded the component of acceleration of a soil movement of duration T along a certain axis by v points. This record (or time series) corresponds to v coordinates (t_i, γ_i) , with the difference $(t_{i+1} - t_i)$ corresponding to the sampling interval τ which is assumed to be practically the same (Fig. 3). Recorded values of acceleration γ_i differ from the real ones

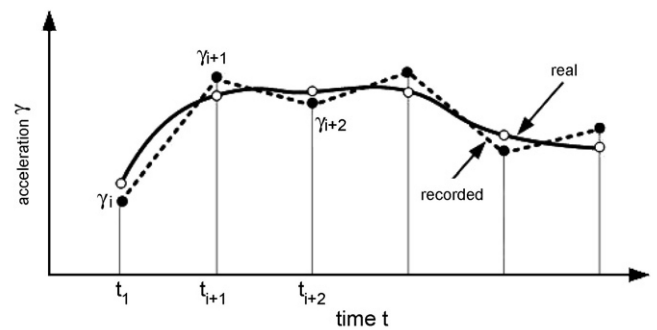


Fig. 3. Differences between real and recorded acceleration γ versus time t .

(Fig. 3), and statistics of their differences $\Delta\gamma_i$ are used to define the standard error σ_γ of measurements.

Velocity and displacement can ideally be computed from the curve of real accelerations using a simple or double integration. In the case of discrete measurements of acceleration γ_i , and assuming linear variation between recordings, Berg and Housner [1] proposed algorithms to compute velocity and displacement using arithmetic integrations. Simpler algorithms based on the assumption of zero initial conditions have been presented by Yang et al. [3] (assuming for simplicity zero acceleration and velocity at $t = 0$; in a different case, initial values of v_0 and of u_0 should be used) and complete algorithms are presented below.

Our approach is based on the assumption that differences $\Delta\gamma_i$ between (unknown) real and measured values γ_i , i.e. measurement errors, are small and random, a reasonable assumption documented below. Consequently, the time series of recorded displacements at a high rate of sampling (i.e., for values of v large enough to permit convergence) approximates the real seismic movement. Hence velocity v_v at time t_v equals to the area between the graph of recorded accelerations γ_i and the horizontal axis between the origin and t_v (Fig. 3) [1,3].

This area can be regarded as the sum of v stripes of infinitesimal width τ , each one bisected by the axis

corresponding to the recorded value γ_i and it can be computed using the formula

$$v_v = \sum_{i=1}^v \tau \gamma_i. \quad (1)$$

The reason to use this formula is that the area δv_v between the curves of real and measured values is practically null, since it corresponds to the sum

$$\delta v_v = \sum_{i=1}^v \tau \Delta \gamma_i = \tau \sum_{i=1}^v \Delta \gamma_i = \tau \cdot 0 = 0, \quad (2)$$

with $\Delta \gamma_i$ being assumed random variables [13]. If systematic errors exist, they may be assumed constant for each record. Such errors can be estimated from experiments and subsequently be used to correct each measurement γ_i .

Computed velocities v_i can be plotted as functions of time and with the same rationale as above, displacement u_v at time t_v can be obtained from the formula

$$u_v = \sum_{i=1}^v \tau v_i. \quad (3)$$

Using Eq. (1) the above formula can be further analyzed to express displacement in terms of observed accelerations, i.e.,

$$\begin{aligned} u_v &= \sum_{i=1}^v \tau v_i = \sum_{i=1}^v \tau (v_1 + v_2 + \dots + v_v) \\ &= \tau [\tau \gamma_1 + \tau (\gamma_1 + \gamma_2) + \dots + \tau (\gamma_1 + \gamma_2 + \dots + \gamma_v)] \\ &= \tau^2 [v \gamma_1 + (v-1) \gamma_2 + \dots + 1 \cdot \gamma_v]. \end{aligned} \quad (4)$$

Thus, Eq. (4) can lead to the formula

$$u_v = \tau^2 \sum_{i=1}^v (v+1-i) \gamma_i. \quad (5)$$

Formulae (1) and (5) permit a precise and simple numerical integration approach to obtain velocity and displacement at time t_v assuming for simplicity and with no loss of generality zero initial conditions ($\gamma_0 = 0$ and $v_0 = 0$ at t_0). In a different case, initial values of v_0 and of u_0 should be added.

2.1. Error properties of accelerometer records

Recorded values of γ_i differ from real values by $\Delta \gamma_i$ because of various errors (Fig. 2). These errors are not a priori known, but an estimate of the mean error σ_γ can be obtained on the basis of various experimental techniques, for instance from a comparison of recorded values by other instruments [8], and especially through experiments with oscillations of known characteristics [14–17]. This last technique permits to define the accuracy, and not simply the precision of an instrument or a technique [15–17]. There is, indeed, much difference between these two terms: precision, described by a standard deviation, defines the dispersion of sampled values from their mean value; on the contrary, accuracy indicates the difference between the

mean value of a sample from the “real” value, known from independent evidence and is significant in the case of non-random (systematic) errors [13,15–17].

Typical (standard) error σ_γ is characteristic of each instrument, method of collection and transformation of the data to be analyzed (for instance analog-to-digital conversion) and also depends on the parameters as well as on the specifications or conditions of each experiment; e.g., a certain instrument may produce higher measurement noise (errors) in the case of higher frequency recordings (for instance [17]), in extreme environmental conditions, etc. Another property of errors in various automatically measuring instruments is that sampling in relatively short records (usually up to a few minutes or hours long) is made under the same environmental and other conditions and the same sampling parameters, and hence all measurements in each recording can be assigned the same weight and hence the same error σ_γ .

A final point concerning errors is that, since all measurements in a certain record come from the same instrument, they are not statistically independent. However, a very reasonable assumption is that in modern instruments all such measurements are very weakly correlated, i.e. the corresponding covariances are practically zero. If this was not true, no acceleration pulses, for instance, could have been identified during earthquakes, and each measurement would strongly depend on previous ones, as if a Kalman or a moving average filter, etc. had been applied to the real recordings. Signal analysis has indeed shown no evidence of important correlations in accelerometer records [9,10], an indication of limited, if any, systematic (non-random) errors.

Automatically recording instruments are contaminated by errors related to the instability of the sampling rate (i.e. variations in the sampling intervals τ). For some instruments indeed, it has been demonstrated that instability in their sampling rate is significant; that it increases with the rate of sampling and may become very important at the upper boundary of their operation [17]. The corresponding error is characteristic not only of a specific instrument, but also of the parameters (preset sampling rate etc) and environmental conditions of measurements.

Instabilities in the sampling rate can be detected on the basis of comparison with a calibration instrument (i.e. an instrument of higher quality, recording at much higher sampling rates), and the corresponding standard error σ_τ in the sampling intervals τ can be defined.

2.2. The law of error propagation for non-correlated variables

The law of propagation of errors in its general form involving Jacobian matrices can be applied to any function [13]. However, since we have assumed that observations of acceleration from a certain record can be practically considered uncorrelated, we may apply the law of error propagation in its simplified form in formulae (4) and (5).

According to this law, the typical error σ_f of a variable $f = f(\chi_1, \chi_2, \dots, \chi_v)$ which is a function of v uncorrelated variables $\chi_1, \chi_2, \dots, \chi_v$ with typical errors $\sigma_{\chi_1}, \sigma_{\chi_2}, \dots, \sigma_{\chi_v}$, is given by the simple formula [13]

$$\sigma_f^2 = \sum_{i=1}^v \left(\frac{\partial f}{\partial \chi_i} \right)^2 \sigma_{\chi_i}^2. \quad (6)$$

2.3. Accelerometer errors assuming a constant sampling rate

At a first step we assume that the sampling interval τ is practically constant, i.e. that the corresponding standard error σ_τ is negligible. Applying formula (6) to Eq. (1) for v uncorrelated variables γ_i which have a common typical error σ_γ (see above), we obtain

$$\sigma_v^2 = \sum_{i=1}^n \left(\frac{\partial v}{\partial \gamma_i} \right)^2 \sigma_\gamma^2 = \sum_{i=1}^v \tau^2 \sigma_\gamma^2 = v \tau^2 \sigma_\gamma^2. \quad (7)$$

Since $T = \tau v$, the above equation yields

$$\sigma_v = \frac{T \sigma_\gamma}{\sqrt{v}}. \quad (8)$$

Similarly, we apply formula (5) to Eq. (4) assuming the same conditions and obtain

$$\sigma_u^2 = \sum_{i=1}^v \left(\frac{\partial u}{\partial \gamma_i} \right)^2, \quad \sigma_\gamma^2 = \sum_{i=1}^v (\tau^2 (v+1-i))^2$$

$$\sigma_\gamma^2 = v \tau^4 \sigma_\gamma^2 \sum_{i=1}^v (v+1-i)^2. \quad (9)$$

Since

$$\sum_{i=1}^v (v+1-i)^2 \approx \frac{1}{3} v^3, \quad (\text{A.1})$$

(as is shown in Appendix 1 for large values of v) and $T = v \tau$, Eq. (9) yields

$$\sigma_u^2 \approx v \tau^4 \sigma_\gamma^2 \frac{1}{3} v^3, \quad (10)$$

or equivalently

$$\sigma_u \approx 0.6 T^2 \sigma_\gamma. \quad (11)$$

2.4. Errors introduced by instabilities in the sampling rate

At a second approximation, we assume that the instability in the sampling interval τ is not negligible and that a specific record we analyze is characterized by a typical error σ_τ , which has been estimated from various experimental techniques. Applying formula (6) to Eqs. (1) and (5) for v measurements γ_i , t_i and for v uncorrelated variables with typical error σ_γ and another variable τ with typical error σ_τ we arrive at the equations

$$\sigma_v^2 = \sum_{i=1}^v \left(\frac{\partial v}{\partial \gamma_i} \right)^2 \sigma_\gamma^2 + \sum_{i=1}^v \left(\frac{\partial v}{\partial \tau} \right)^2 \sigma_\tau^2, \quad (12)$$

$$\sigma_u^2 = \sum_{i=1}^v \left(\frac{\partial u}{\partial \gamma_i} \right)^2 \sigma_\gamma^2 + \sum_{i=1}^v \left(\frac{\partial u}{\partial \tau} \right)^2 \sigma_\tau^2, \quad (13)$$

which yield

$$\sigma_v^2 = \sum_{i=1}^v \tau^2 \sigma_\gamma^2 + \sum_{i=1}^v \gamma_i^2 \sigma_\tau^2 = v \sigma_\gamma^2 \tau^2 + \sigma_\tau^2 \sum_{i=1}^v \gamma_i^2, \quad (14)$$

$$\sigma_u^2 = \sum_{i=1}^v [\tau^2 (v+1-i)]^2 \sigma_\gamma^2 + [2\tau \sum_{i=1}^v (v+1-i) \gamma_i]^2,$$

$$\sigma_\tau^2 \approx \frac{1}{3} v^4 \tau^4 \sigma_\gamma^2 + 4\tau^2 \sigma_\tau^2 \left(\sum_{i=1}^v (v+1-i) \gamma_i \right)^2. \quad (15)$$

If we introduce a variable $\tilde{\gamma}$ such that

$$\tilde{\gamma}^2 = \frac{1}{v} \sum_{i=1}^v \gamma_i^2, \quad (16)$$

on the basis of (A.1), and at a crude approximation, Eq. (14) is simplified to

$$\sigma_u^2 \approx \frac{1}{3} v^4 \tau^4 \sigma_\gamma^2 + 4\tau^2 \sigma_\tau^2 \tilde{\gamma}^2 \frac{1}{3} v^3 \approx \frac{1}{3} T^4 \sigma_\gamma^2 + \frac{4}{3} T^2 v \sigma_\tau^2 \tilde{\gamma}^2. \quad (17)$$

3. Summary

Assuming zero initial conditions, it has been shown that velocity v_t and displacement u_t at time t_v are defined from observations of acceleration γ_i by the equations

$$v_v = \sum_{i=1}^v \tau \gamma_i, \quad (1)$$

$$u_v = \tau^2 \sum_{i=1}^v (v+1-i) \gamma_i, \quad (5)$$

and the corresponding typical errors are defined by the equations

$$\sigma_v = \frac{T \sigma_\gamma}{\sqrt{v}}, \quad (8)$$

$$\sigma_u \approx 0.6 T^2 \sigma_\gamma, \quad (11)$$

if the typical error in the sampling interval is insignificant, or by the equations

$$\sigma_v^2 = \sum_{i=1}^v \tau^2 \sigma_\gamma^2 + \sum_{i=1}^v \gamma_i^2 \sigma_\tau^2 = v \sigma_\gamma^2 \tau^2 + \sigma_\tau^2 \sum_{i=1}^v \gamma_i^2, \quad (14)$$

$$\sigma_u^2 = \sum_{i=1}^v [\tau^2 (v+1-i)]^2 \sigma_\gamma^2 + [2\tau \sum_{i=1}^v (v+1-i) \gamma_i]^2,$$

$$\sigma_\tau^2 \approx \frac{1}{3} v^4 \tau^4 \sigma_\gamma^2 + 4\tau^2 \sigma_\tau^2 \left(\sum_{i=1}^v (v+1-i) \gamma_i \right)^2, \quad (15)$$

if it is significant. Furthermore, Eq. (15) can be written approximately as

$$\sigma_u^2 \approx \frac{1}{3} v^4 \tau^4 \sigma_\gamma^2 + 4 \tau^2 \sigma_\gamma^2 \bar{\gamma}^2 \frac{1}{3} v^3 \approx \frac{1}{3} T^4 \sigma_\gamma^2 + \frac{4}{3} T^2 v \sigma_\gamma^2 \bar{\gamma}^2. \quad (17)$$

4. Discussion

Eq. (8) indicates that the error in velocity derived from acceleration depends on the duration of the record, as well as on the sensitivity/accuracy of the measurements described by σ_γ . The latter probably depends on the instrument as well as on the parameters and environmental conditions of measurements. This error is reduced by an increase in the sampling frequency, as the division by the square root of v indicates. Still, at the upper limits of operation of various instruments, a very rapid increase in errors is expected [17]. The duration T of the signal seems to be the major factor controlling the accuracy of displacements [9] because of its quadratic form in Eq. (11). This explains the important drift in Fig. 2.

Errors in velocity are increased by instabilities in the sampling rate, as Eq. (14) reveals. These errors depend on the sum of squares of recorded accelerations, mainly influenced by high peaks; hence velocity and displacement errors are, to a major degree, controlled by high peaks in the accelerogram record, in agreement with [11]. This is also the case with errors with displacements, as Eqs. (15) and (17) reveal. Eq. (15) in particular shows that early observations are given a high weight, as is clearly shown in Eq. (4). This indicates that it is not only the amplitude of peaks which influence errors, but their timing as well: high peaks at the beginning of the timeseries will have a much important impact in the error budget than later peaks (amplitude multiplied by a larger integer, as is depicted in Eq. (4)).

The above analysis was limited to linear movements and hence rotations were ignored. In reality, accelerographs do not remain vertical, but are subject to time-dependent rotations. Hence, computed velocities and displacements along any axis are contaminated by a sinusoidal noise. Still, this noise is usually of small amplitude, and although important for deriving absolute displacements [11,18], its contribution in the overall error budget is rather limited and may be ignored for most structures; except for large angular oscillations in which, however, the main source of error seems to be the frequency characteristic error of the instrument [12].

A practical question that arises is how errors can be modeled and eliminated. Previous discussion indicates that various methods for correction of accerograph-derived data [2,3,11] may be refined if baseline corrections take into consideration the equations of errors proposed above. Still, errors can be fully estimated only in comparison with other instruments directly determining displacement, for instance GPS [7,16].

Acknowledgments

This note benefited from constructive comments by D. Beskos and two anonymous reviewers.

Appendix 1

Computation of the quantity $\sum_{i=1}^v (v+1-i)^2$
One has

$$\begin{aligned} \sum_{i=1}^v (v+1-i)^2 &= \sum_{i=1}^v (v^2 + 1^2 + i^2 + 2v - 2vi - 2 \times 1i) \\ &= v v^2 + v 1 + \sum_{i=1}^v i^2 + v 2v - 2v \sum_{i=1}^v i - 2 \sum_{i=1}^v i \\ &= v^3 + v + \frac{1}{6} v(v+1)(2v+1) + 2v^2 \\ &\quad - v^2(v+1) - v(v+1) \\ &= \frac{v(2v^2 + 3v + 1)}{6} = \frac{1}{3} v^2 \left(v + \frac{1}{2} + \frac{1}{v} \right), \end{aligned}$$

using the common formulae for sums of infinite series of integers and of squares of integers $\sum_{i=1}^v i = v(v+1)/2$ and $\sum_{i=1}^v i^2 = v(v+1)(2v+1)/6$ since $v \neq 0$.

Furthermore, for large values of v , $(v+1/2+1/v) \rightarrow v$. Hence

$$\sum_{i=1}^v (v+1-i)^2 \approx \frac{1}{3} v^3. \quad (A.1)$$

References

- [1] Berg GV, Housner GW. Integrated velocity and displacement of strong earthquake ground motion. *Bull Seism Soc Am* 1961;51(2): 175–89.
- [2] Boore DM, Bommer JJ. Processing of strong-motion accelerograms: needs, options and consequences. *Soil Dyn Earthquake Eng* 2005;25: 93–115.
- [3] Yang J, Li J, Lin G. A simple approach to integration of acceleration data for dynamic soil–structure interaction analysis. *Soil Dyn Earthquake Eng* 2006;26:725–34.
- [4] Hide C, Blake S, Meng X, Roberts G, Moore T, Park D. An investigation in the use of GPS and INS sensors for structural health monitoring. *Proc.18th Int. Technical Meeting of the Satellite Division of The Institute of Navigation, ION GNSS 2005*: 2029–38.
- [5] Lee HK, Wang J, Rizos C, Grejner-Brzezinska D. Analyzing the impact of integrating pseudolite observables into a GPS/INS system. *J Surveying Eng* 2004;130:95–103.
- [6] Semeniuk L, Noureldin A. Bridging GPS outages using neural network estimates of INS position and velocity errors. *Meas Sci Technol* 2006;17:2783–98.
- [7] Wang GQ, Boore DM, Igel H, Zhou XY. Some observations on collocated and closely spaced strong ground-motion records of the 1999 Chi-Chi, Taiwan, earthquake. *Bull Seism Soc Am* 2003;93(2): 674–93.
- [8] Trifunac M, Lee V. A note on the accuracy of computed ground displacements from strong-motion accelerograms. *Bull Seism Soc Am* 1974;64:1209–19.
- [9] Thong Y, Woolfson M, Crowe J, Hayes-Gill B, Challis R. Dependence of inertial measurements of distance on accelerometer noise. *Meas Sci Technol* 2002;13:1163–72.

- [10] Thong Y, Woolfson M, Crowe J, Hayes-Gill B, Jones D. Numerical double integration of acceleration measurements in noise. *Meas Sci Technol* 2004;36:73–92.
- [11] Trifunac M, Todorovska M. A note on the useable dynamic range of accelerographs recording translation. *Soil Dyn Earthquake Eng* 2001;21:275–86.
- [12] Hu H, Ma J. Error analysis on linear accelerometer-assisted inertial angular position measurement. *Syst Eng Electron* 2006;28:284–7 [in Chinese].
- [13] Mikhail E. *Observations and least squares*. New York: IEP-A Dun-Donnelley Publisher; 1976.
- [14] Celebi M, Sanli A. GPS in pioneering dynamic monitoring of long-period structures. *Earthquake Spectra* 2002;18:47–61.
- [15] Nickitopoulou A, Protosalti K, Stiros S. Monitoring dynamic and quasi-static deformations of large flexible engineering structures with GPS: accuracy, limitations and promises. *Eng Struct* 2006;28:1471–82.
- [16] Chan WS, XU YL, Ding XL, Xiong XL. Assessment of dynamic measurement accuracy of GPS in three directions. *J Surv Eng* 2006;132(3):108–17.
- [17] Psimoulis P, Stiros S. Monitoring and spectral characteristics of flexible engineering structures using robotic theodolite (RTS). *Engineering structures*, November 2006 (submitted).
- [18] Graizer VM. Effect of tilt on strong motion data processing. *Soil Dyn Earthquake Eng* 2005;25:197–204.